> $int\left(\frac{t^m}{\mathrm{sqrt}(1-t^2)}, t=a..b\right)$ assuming $m :: posint$

$$-\frac{1}{1+m}\left(a^{1+m}\,\mathrm{hypergeom}\left(\left[\frac{1}{2}, \frac{1}{2}+\frac{m}{2}\right], \left[\frac{3}{2}+\frac{m}{2}\right], a^2\right) - b^{1+m}\,\mathrm{hypergeom}\left(\left[\frac{1}{2}, \frac{1}{2}+\frac{m}{2}\right], \left[\frac{3}{2}+\frac{m}{2}\right], b^2\right)\right) \quad \textbf{(1)}$$

> $simplify(\mathbf{(1)})$ assuming $m :: posint, -1 < a < b < 1$

$$\frac{1}{1+m}\left(-a^{1+m}\,\mathrm{hypergeom}\left(\left[\frac{1}{2}, \frac{1}{2}+\frac{m}{2}\right], \left[\frac{3}{2}+\frac{m}{2}\right], a^2\right) + b^{1+m}\,\mathrm{hypergeom}\left(\left[\frac{1}{2}, \frac{1}{2}+\frac{m}{2}\right], \left[\frac{3}{2}+\frac{m}{2}\right], b^2\right)\right) \quad \textbf{(2)}$$

> $map(simplify, [seq(eval(eval(\mathbf{(2)}, a=-1), b=1), m=1..20)])$

$$\left[0, \frac{\pi}{2}, 0, \frac{3\pi}{8}, 0, \frac{5\pi}{16}, 0, \frac{35\pi}{128}, 0, \frac{63\pi}{256}, 0, \frac{231\pi}{1024}, 0, \frac{429\pi}{2048}, 0, \frac{6435\pi}{32768}, 0, \frac{12155\pi}{65536}, 0, \frac{46189\pi}{262144}\right] \quad \textbf{(3)}$$

> # *https://oeis.org/A001790 # A001790 Numerators in expansion of 1/sqrt(1-x). - This sequence gives the numerators of the* ***Maclaurin series of the Lorentz factor*** *(see Wikipedia link) of* ***1/sqrt(1-b^2) = dt/dtau*** *where* ***b=u/c*** *is the velocity in terms of the speed of light c, u is the velocity as observed in the reference frame where time t is measured and tau is the proper time. — Stephen Crowley, Apr 03 2007 # 1, 1, 3, 5, 35, 63, 231, 429, 6435, 12155, 46189, 88179,*

> $today := Date(\ );$

$$today := \langle Date: 2023\text{-}09\text{-}23T21:51:24.912\ GMT\rangle \quad \textbf{(4)}$$

> $u := t \rightarrow \frac{1}{\mathrm{sqrt}(1-t^2)}$

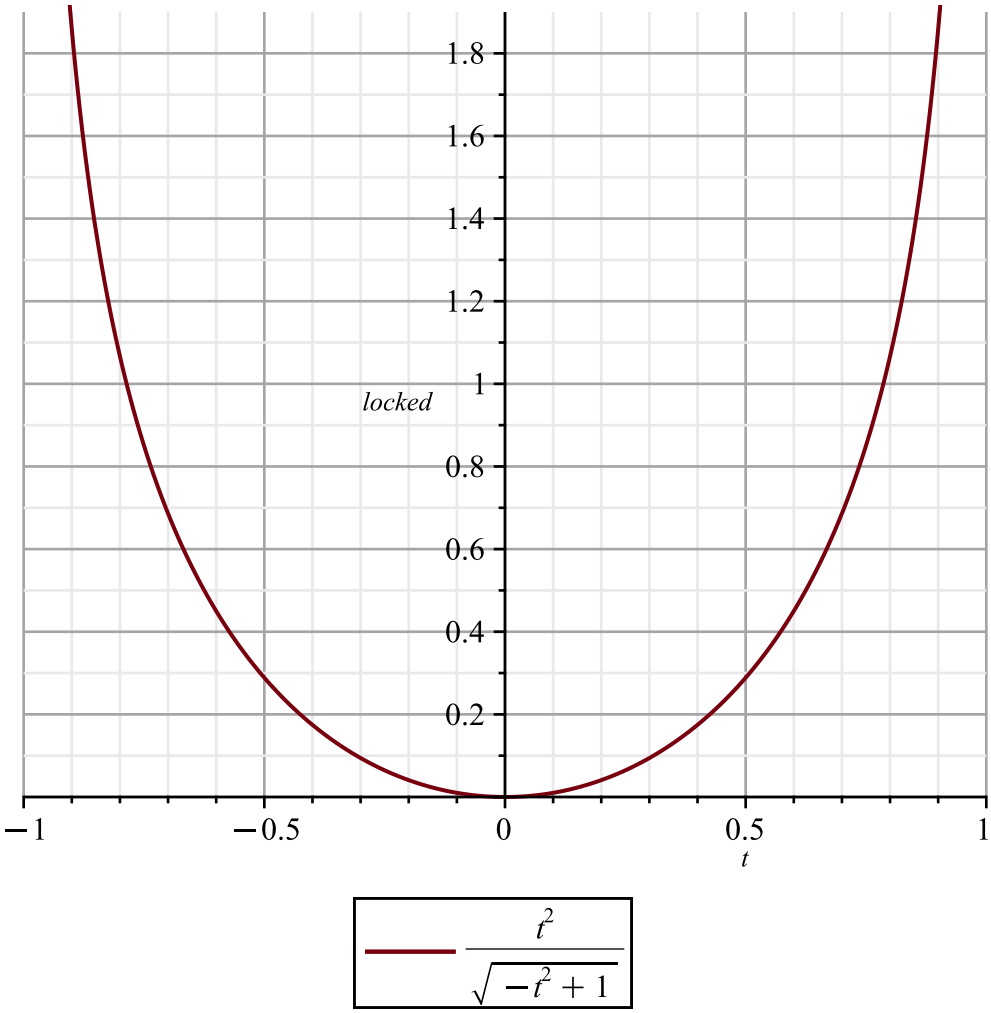
$$u := t \mapsto \frac{1}{\sqrt{1-t^2}} \quad \textbf{(5)}$$

> $int(t^2 \cdot u(t), t=-1..1);$

 # The integral converges because the singularites are on the boundaries of the domain (-1,1) rather than the interior of ithprime

$$\frac{\pi}{2} \quad \textbf{(6)}$$

> $plot(t^2 \cdot u(t), t=-1..1, gridlines=true, locked, legend=[t^2 \cdot u(t)])$


1.8
1.6
1.4
1.2
1
locked
0.8
0.6
0.4
0.2
−1
−0.5
0
0.5
1
t
$\frac{t^2}{\sqrt{-t^2+1}}$

>